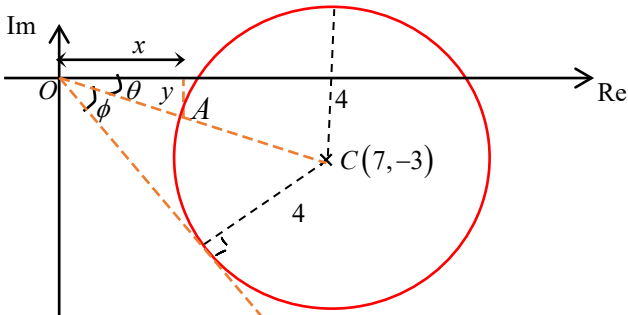
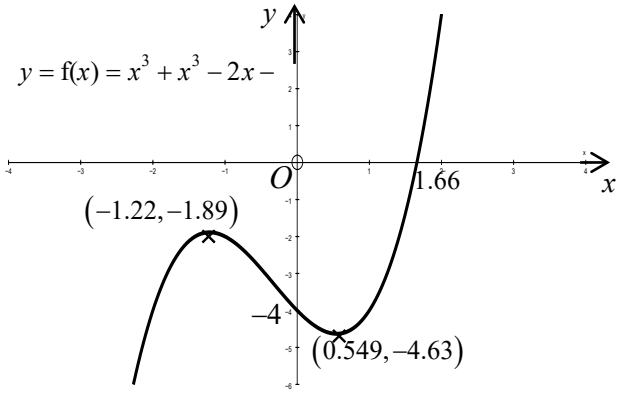
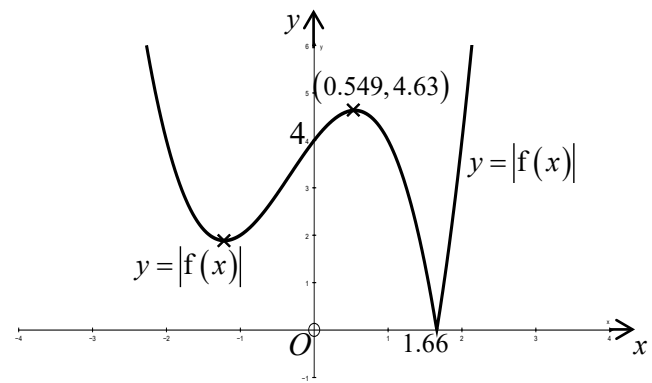
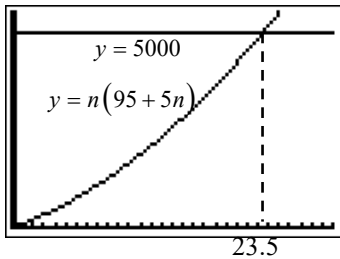
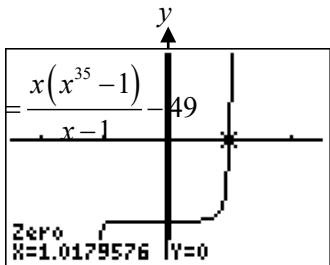


Section A: Pure Mathematics

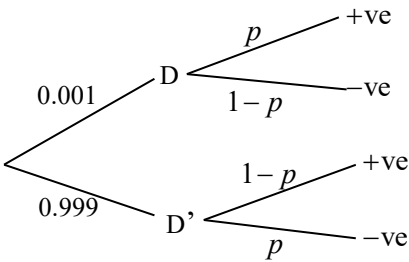
No.	Solutions	Comments
1a	$\frac{d^2y}{dx^2} = 16 - 9x^2$ $\frac{dy}{dx} = 16x - 3x^3 + C$ $y = 8x^2 - \frac{3}{4}x^4 + Cx + D$	Remember to add in the arbitrary constants. You should know that every time an indefinite integration is done, it will give rise to an arbitrary constant. A second order DE will have 2 arbitrary constants in the general solution.
1b	$\frac{du}{dt} = 16 - 9u^2 \Rightarrow \int \frac{1}{16 - 9u^2} du = \int dt$ $\Rightarrow \frac{1}{9} \int \frac{1}{\left(\frac{4}{3}\right)^2 - u^2} du = \int dt$ $\Rightarrow \frac{1}{9} \cdot \frac{1}{2\left(\frac{4}{3}\right)} \ln \left \frac{\frac{4}{3} + u}{\frac{4}{3} - u} \right = t + C$ $\Rightarrow \frac{1}{24} \ln \left \frac{4 + 3u}{4 - 3u} \right - C = t$ When $t = 0$, $u = 1 \Rightarrow \frac{1}{24} \ln 7 - C = 0 \therefore C = \frac{1}{24} \ln 7$ $\therefore t = \frac{1}{24} \ln \left \frac{4 + 3u}{7(4 - 3u)} \right $	It is faster to use MF26 to integrate, compared to using partial fractions. Take note that full credit was only given if you include the constant in a single logarithmic form for t in terms of u.
2i		Not in H2 Math 9758 syllabus Be careful when marking out the centre of the circle. For a circle, both the coordinates of the centre and the radius have to be marked out.
2iia	$ z = OA = OC - AC = \sqrt{7^2 + 3^2} - 4 = \sqrt{58} - 4$	Not in H2 Math 9758 syllabus
2iib	$x = OA \cos \theta = 7 \left(\frac{\sqrt{58} - 4}{\sqrt{58}} \right), y = OA \sin \theta = 3 \left(\frac{\sqrt{58} - 4}{\sqrt{58}} \right)$ $\therefore z = x - iy = \frac{7(\sqrt{58} - 4)}{\sqrt{58}} - \frac{3(\sqrt{58} - 4)}{\sqrt{58}} i$	Not in H2 Math 9758 syllabus Note that exact answer is required. Using trigonometric ratio and obtaining $\theta = \tan^{-1} \left(-\frac{3}{7} \right)$ then either giving an approximate decimal answer or giving an answer in the form $re^{i\theta}$ will not gain full credit.

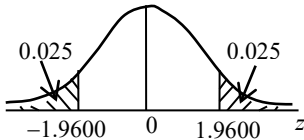
2iii	$\theta = \tan^{-1} \frac{3}{7}, \phi = \sin^{-1} \frac{4}{\sqrt{58}}$ $\therefore \arg z = -(\theta + \phi) = -0.9579 \text{ (4 s.f.)}$	Not in H2 Math 9758 syllabus The most common mistake was the omission of the minus sign. Remember that direction (clockwise or anti-clockwise) has to be taken into account when writing down $\arg z$.
3i	 $y = f(x) = x^3 + x^2 - 2x - 4$	All the essential features of a graph need to be indicated e.g. the coordinates of the stationary points, x-and y- intercepts.
3ii	$x^3 + x^2 - 2x - 4 = 4$ $x^3 + x^2 - 2x - 8 = 0$ $(2)^3 + (2)^2 - 2(2) - 8 = 8 + 4 - 4 - 8 = 0$ By factor theorem, $x = 2$ is a solution. $\therefore (x - 2)(x^2 + 3x + 4) = 0$ Discriminant of $x^2 + 3x + 4 = 3^2 - 4(4) = -7 < 0$ $\therefore x^2 + 3x + 4 = 0 \text{ has no real solution.}$ $\therefore x = 2 \text{ is the only real solution}$	The proving of the second part seems to be difficult for a large number of students. Note that there is a difference between $x = 2$ (which is a solution) and $(x - 2)$ which is a factor.
3iii	Solution is $x + 3 = 2 \Rightarrow x = -1$	The most efficient way was to make use of (ii) to solve this part. Of course using a GC to solve the equation from scratch will still be given the full credit but definitely much more time consuming.
3iv	 $y = f(x)$	Again essential features marked out in (i) has to be indicated here. Take note of the shape of the curve at $(1.66, 0)$ - not a stationary point.
3v	$ f(x) = 4 \Rightarrow f(x) = \pm 4$	Note down the need to write down the 2 different cubic equations. There is a difference between $y = x^3 + x^2 - 2x - 8$ and $x^3 + x^2 - 2x - 8 = 0$. Be

	$x^3 + x^2 - 2x - 8 = 0 \dots\dots\dots (1)$ $-x^3 - x^2 + 2x = 0 \dots\dots\dots (2)$ $(1) \Rightarrow x = 2$ $(2) \Rightarrow x^3 + x^2 - 2x = 0$ $\Rightarrow x(x^2 + x - 2) = 0$ $\Rightarrow x(x - 1)(x + 2) = 0$ $\Rightarrow x = 0 \text{ or } 1 \text{ or } -2$ $\therefore \text{ roots of } f(x) = 4 \text{ are } 0, 1, \pm 2$	careful not to omit the root $x = 0$. By looking at the graph of $y = f(x) $, you should expect 4 roots.																		
4i	100, 110, 120, AP, $a = 100, d = 10$ $S_n > 5000 \Rightarrow \frac{n}{2}[2 \times 100 + (n - 1)10] > 5000$ $\therefore n(95 + 5n) > 5000$ Method 1 <table border="1"><tr><td>n</td><td>$S_n = n(95 + 5n)$</td></tr><tr><td>23</td><td>$4830 < 5000$</td></tr><tr><td>24</td><td>$5160 > 5000$</td></tr></table> Method 2  $\therefore n \geq 24$ Mrs A's account first become $> \\$5000$ on 1 Dec 2002.	n	$S_n = n(95 + 5n)$	23	$4830 < 5000$	24	$5160 > 5000$	Most students were able to arrive at the answer $n \geq 24$, however many students correct up to this point were not able to write down the date correctly. You may want to try listing a few terms so as to detect a pattern which may help arrive at the correct date.												
n	$S_n = n(95 + 5n)$																			
23	$4830 < 5000$																			
24	$5160 > 5000$																			
4ii	<table border="1"><tr><td>n</td><td>Amount on 1st day of month n</td><td>Amount at end of month n</td></tr><tr><td>1</td><td>100</td><td>$1.005(100)$</td></tr><tr><td>2</td><td>$100 + 1.005(100)$ $= 100(1 + 1.005)$</td><td>$1.005[100(1 + 1.005)]$ $= 100(1.005 + 1.005^2)$</td></tr><tr><td>3</td><td>$100 + 100(1.005 + 1.005^2)$ $= 100(1 + 1.005 + 1.005^2)$</td><td>$1.005[100(1 + 1.005 + 1.005^2)]$ $= 100(1.005 + 1.005^2 + 1.005^3)$</td></tr><tr><td>$\vdots$</td><td>$\vdots$</td><td>$\vdots$</td></tr><tr><td>$n$</td><td>$100(1 + 1.005 + \dots + 1.005^{n-1})$</td><td>$100(1.005 + \dots + 1.005^n)$</td></tr></table> $\text{Amount at end of } n^{\text{th}} \text{ month} = \$ \frac{100(1.005)(1 - 1.005^n)}{1 - 1.005}$ $= \$20100(1.005^n - 1)$	n	Amount on 1 st day of month n	Amount at end of month n	1	100	$1.005(100)$	2	$100 + 1.005(100)$ $= 100(1 + 1.005)$	$1.005[100(1 + 1.005)]$ $= 100(1.005 + 1.005^2)$	3	$100 + 100(1.005 + 1.005^2)$ $= 100(1 + 1.005 + 1.005^2)$	$1.005[100(1 + 1.005 + 1.005^2)]$ $= 100(1.005 + 1.005^2 + 1.005^3)$	$\vdots$	$\vdots$	$\vdots$	n	$100(1 + 1.005 + \dots + 1.005^{n-1})$	$100(1.005 + \dots + 1.005^n)$	This part was badly done. Some of the common mistakes include using 0.005, 1.05, 1.5 as the common ratio. The first term of the GP was also incorrectly given as \$100. There are other students who did not manage to convert their value of n into the right month and year. Note that amount at the end of 44th month (end of Aug 2004) is \$4932.42. So at the start of Sep 2004, the amount (in dollars) would be $4932.42 + 100 = 5032.42$.
n	Amount on 1 st day of month n	Amount at end of month n																		
1	100	$1.005(100)$																		
2	$100 + 1.005(100)$ $= 100(1 + 1.005)$	$1.005[100(1 + 1.005)]$ $= 100(1.005 + 1.005^2)$																		
3	$100 + 100(1.005 + 1.005^2)$ $= 100(1 + 1.005 + 1.005^2)$	$1.005[100(1 + 1.005 + 1.005^2)]$ $= 100(1.005 + 1.005^2 + 1.005^3)$																		
$\vdots$	$\vdots$	$\vdots$																		
n	$100(1 + 1.005 + \dots + 1.005^{n-1})$	$100(1.005 + \dots + 1.005^n)$																		

	$\therefore 20100(1.005^n - 1) > 5000$ Method 1: <table border="1"><tr><td>n</td><td>$20100(1.005^n - 1)$</td></tr><tr><td>44</td><td>$4932.42 < 5000$</td></tr><tr><td>45</td><td>$5057.58 > 5000$</td></tr><tr><td>$\vdots$</td><td>$\vdots$</td></tr></table> Method 2 $20100(1.005^n - 1) > 5000$ $1.005^n > \frac{5000}{20100} + 1$ $n \ln(1.005) > \ln\left(\frac{5000}{20100} + 1\right)$ $n > 44.541$ $n \geq 45 \text{ which is minimum 3 years and 9 months}$ Mr B's account first become $> \\$5000$ at the start of 45th month i.e. first day of Sept 2004	n	$20100(1.005^n - 1)$	44	$4932.42 < 5000$	45	$5057.58 > 5000$	$\vdots$	$\vdots$	
n	$20100(1.005^n - 1)$									
44	$4932.42 < 5000$									
45	$5057.58 > 5000$									
$\vdots$	$\vdots$									
4iii	\$5000 on 2 Dec 2003 $\therefore \frac{100x(1-x^{35})}{1-x} + 100 = 5000 \Rightarrow \frac{x(x^{35}-1)}{x-1} - 49 = 0$ From GC: $x = 1.01796$ $\therefore$ interest rate of 1.80 % 	This part proved difficult for a lot of students. Students failed to recognise that since the amount in the account on 2 Dec 2003 is required, there is a need to + \$100 for the amount put in on 1 Dec 2003. Do remember to convert common ratio found into monthly percentage interest rate. $b\%$ - monthly interest rate $\Rightarrow$ $1 + \frac{b}{100} = 1.01796$								

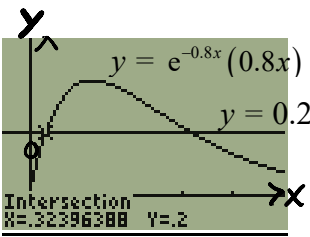
Section B: Statistics

No.	Solutions	Comments
5		
5ia	$p = 0.995$ $P(\text{test result is positive}) = 0.001p + 0.999(1-p)$ $= 0.999 - 0.998 \times 0.995 = 0.00599$	Make use of a tree diagram so as not to miss out cases.
5ib	$P(\text{patient has the disease} \text{test result is positive})$	

	$= \frac{P(\text{patient has the disease and test result is positive})}{P(\text{test result is positive})}$ $= \frac{0.001 \times 0.995}{0.00599} \approx 0.166$	
5ii	$P(\text{patient has the disease} \text{test result is positive}) = 0.75$ $\therefore \frac{0.001p}{0.001p + 0.999(1-p)} = 0.75$ $\Rightarrow 0.001p = 0.75(0.999 - 0.998p) \Rightarrow p \approx 0.999666$	Read the question carefully. Leave your answer correct to the required decimal places.
6i	Let X cm be the tail length of a squirrel on the island and μ cm be the population mean. $H_0 : \mu = 14.0$ $H_1 : \mu \neq 14.0$	Make it a habit to define the random variable.
6ii	Level of significance: 5% Test statistic: When H_0 is true, $Z = \frac{\bar{X} - 14.0}{\frac{3.8}{\sqrt{n}}} \sim N(0,1)$ Rejection region: $z \leq -1.9600$ or $z \geq 1.9600$ Computation: $z = \frac{\bar{x} - 14.0}{3.8/\sqrt{20}}$  H_0 is not rejected if $-1.9600 < z < 1.9600$ $\Rightarrow -1.9600 < \frac{\bar{x} - 14.0}{3.8/\sqrt{20}} < 1.9600$ $\Rightarrow 14.0 - 1.9600 \times \frac{3.8}{\sqrt{20}} < \bar{x} < 14.0 + 1.9600 \times \frac{3.8}{\sqrt{20}}$ $\Rightarrow 12.3 < \bar{x} < 15.7$ Required set is $\{\bar{x} \in \mathbb{R} : 12.3 < \bar{x} < 15.7\}$	Since $\bar{x}$ is not known, we will not be able to calculate p-value. Hence rejection region has to be found. Note that for a hypothesis test, we compare either 1) p-value against level of significance or 2) standardised $\bar{x}$ value against critical values which is $\text{invNorm}(0.025)$ and $-\text{invNorm}(0.025)$.
6iii	$\bar{x} = 15.8$ is not within $12.3 < \bar{x} < 15.7$, $\therefore H_0$ is rejected at 5% significance level. Hence there is sufficient evidence to conclude that squirrels on the island do not have the same mean tail length as the species known to the zoologist (i.e. 14.0 cm).	Use part (ii). Give a conclusion in the context of the question.
7i	$P(\text{the sisters are next to each other}) = \frac{14!2!}{15!} = \frac{2}{15}$	Read the question properly - you want to calculate the probability, not the number of ways in which the event can happen.

7ii	Req. prob. = $1 - P(\text{the brothers are all next to each other})$ $= 1 - \frac{13!3!}{15!} = \frac{34}{35}$	
7iii	Req. prob. = $\frac{12!2!3!}{15!} = \frac{2}{455}$ S_1S_2 $B_1B_2B_3$ 10 others	You should not be doing $P(A \cap B) = P(A) \times P(B)$ since you cannot assume that the 2 events are independent.
7iv	Let S and B denote the events 'sisters are next to each other' and 'brothers are all next to one another' respectively. $P(S \cup B) = P(S) + P(B) - P(S \cap B)$ $= \frac{2}{15} + \frac{1}{35} - \frac{2}{455} = \frac{43}{273}$	Think of this part as $P(A \cup B)$ and (i) is $P(A)$, (ii) is $P(B)$ while (iii) is $P(A \cap B)$
7v	$P(\text{the sisters are next to each other}) = \frac{1.13!2!}{1.14!} = \frac{1}{7}$	
8i		x must be placed on the horizontal axis as it is the variable that is "controlled".
8ii	Week 5's mark does not seem to follow the trend of marks (improving with time). It could be due to a harder paper in week 5.	
8iii	A linear model is not appropriate as it would lead to y exceeding 100% eventually, and that is not possible (or as the points in the scatter diagram do not seem to lie close to a straight line, a linear model is not appropriate). A quadratic model is also not appropriate as the graph would eventually reach a maximum and then decrease with increasing amounts, which is not likely to happen in this context.	You need to comment on both linear and quadratic model in your answer before you can get the full marks.
8iv	$r \approx -0.92974412 \approx -0.929744$	remember to give your answer correct to 6 d.p.
8v	92 is the most appropriate value for L since r is closest to -1 when $L = 92$.	must give a reason to support your choice.
8vi	The regression line of $\ln(92 - y)$ on x is given by $\ln(92 - y) \approx 4.1045 - 0.27960x$ $\therefore a \approx 4.10, b \approx -0.280$ For $90 \leq y < 92$, $\ln(92 - y) \leq \ln 2$ $\Rightarrow 4.1045 - 0.27960x \leq \ln 2 \Rightarrow x \geq 12.2$	Alternatively, $\ln(92 - y) \approx 4.1045 - 0.27960x$ $y = 92 - e^{4.1045 - 0.27960x}$

	Week 13 is the required week.	$y \geq 90$ $92 - e^{4.1045 - 0.27960x} \geq 90$ $4.1045 - 0.27960x \leq \ln 2$ $x \geq 12.2$
8vii	The value of L is Amy's likely long term maximum mark if she continues to practice indefinitely i.e. $y \rightarrow L$ as $x \rightarrow \infty$	$\ln(L - y) = a + bx$ $\Rightarrow L - y = e^{a+bx}$ $y = L - e^{4.10 - 0.280x}$ All these answers are wrong: max mark anyone could get, total mark for the paper, mark Amy would like to get.
9i	For A to be well modelled by a binomial distribution, the probability that a voter supports the Alliance Party remains constant and that the opinions of the voters (i.e. support or not support the Party) are independent of each other .	You should NOT say "number of voters are independent". "probabilities are independent":
9ii	$A \sim B(30, 0.15)$ $P(A = 3 \text{ or } 4) = P(A = 3) + P(A = 4) = 0.37307 \approx 0.373$	The word "or" in Mathematics means "union". So we add the 2 probabilities up.
9iia	$A \sim B(30, 0.55)$ Since $n = 30$ is sufficiently large such that $np = 30 \times 0.55 = 16.5 > 5$ and $n(1 - p) = 30 \times 0.45 = 13.5 > 5$, it is possible to approximate the distribution of A with a normal distribution.	Not in H2 Math 9758 syllabus You need to give reasons for choosing normal distribution as an approximation of A AND rejecting Poisson distribution as an approximation of A .
9iiib	As $np = 30 \times 0.55 = 16.5 > 5$, it is not possible to approximate the distribution of A with a Poisson distribution.	Not in H2 Math 9758 syllabus
9iv	$A \sim B(30, p)$ $P(A = 15) = 0.06864 \Rightarrow \binom{30}{15} p^{15} (1 - p)^{15} = 0.06864$ $p(1 - p) = \left[\frac{0.06864}{155117520} \right]^{\frac{1}{15}} = 0.23790$ $p^2 - p + 0.23790 = 0$ $\Rightarrow p = 0.39$ or $p = 0.61$ Since $p < 0.5$, $p = 0.39$	
10i	Two conditions: (1) Gold coins are scattered independently. (2) The average number of gold coins scattered in a one square metre of space is constant.	Not in H2 Math 9758 syllabus "Gold coins are scattered randomly" is also acceptable.
10ii	Let X be the number of gold coins in a region of 1 square metre. $X \sim P_0(0.8)$ $P(X \geq 3) = 1 - P(X \leq 2) = 0.0474$	Not in H2 Math 9758 syllabus
10iii	Let Y be the number of gold coins in a region of x square	Not in H2 Math 9758 syllabus

	metres. $Y \sim P_0(0.8x)$  $P(Y=1) = 0.2$ $\Rightarrow e^{-0.8x}(0.8x) = 0.2$ From GC: $x = 0.324$ or 3.18 Since $x < 1$, $x = 0.324$.	
10iv	Let W be the number of gold coins in a region of 100 square metres. $W \sim P_0(\lambda), \lambda = 100 \times 0.8 = 80$ Since $\lambda = 80 > 10$, use normal approximation. $E(W) = \text{Var}(W) = 80$ $\therefore W \sim N(80, 80) \text{ approximately}$ $P(W \geq 90) = P(W > 89.5) \approx 0.14409 \approx 0.144$	Not in H2 Math 9758 syllabus Use continuity correction and check backwards to make sure your correction is done correctly.
10v	Let U and V be the number of gold coins and pottery shards in a region of 50 square metres respectively. $U \sim P_0(50 \times 0.8), V \sim P_0(50 \times 3)$ $U + V \sim P_0(40 + 150) \text{ i.e. } U + V \sim P_0(190)$ Since $\lambda = 190 > 10$, $U + V \sim N(190, 190)$ approximately. $P(U + V \geq 200) = P(U + V \geq 199.5) \approx 0.24535 \approx 0.245$	Not in H2 Math 9758 syllabus If we are dealing with $U - V$, then we will need to approximate U and V separately to normal distributions first before we can approximate $U - V$ to a normal distribution.
10vi	$U \sim N(40, 40) \text{ approximately}$ $V \sim N(150, 150) \text{ approximately}$ To find $P(V \geq 3U) = P(V - 3U \geq 0)$ $E(V - 3U) = E(V) - 3E(U) = 150 - 3 \times 40 = 30$ $\text{Var}(V - 3U) = \text{Var}(V) + 9\text{Var}(U) = 150 + 9 \times 40 = 510$ $V - 3U \sim N(30, 510) \text{ approximately}$ $\therefore P(V - 3U \geq 0) = P(V - 3U > -0.5) \approx 0.91158 \approx 0.912$	Not in H2 Math 9758 syllabus